

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2054

Hawkins 138 kV
April 28, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2054. The Interconnection Customer proposes a 162.4 MW solar plus storage facility interconnecting at the Hawkins 138 kV substation in the New Jersey area. The request was received on April 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,289,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Florida Avenue to Pine Hill 138 KV	Msc 345-138 kV Transformer Bank	129.7	212	39.8
Hawkins Substation to Pine Hill Substation 138 KV	Msc Substation 345-138 kV Transformer	121.1	240	59.8
Florida Avenue to Bison 138 KV	Msc 345-138 kV TX	106.2	193	39.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pine Hill to Florida Avenue 138 KV	Rebuild line with higher-capacity conductor	\$9,628,000	73% (\$7,061,000)
Hawkins to Pine Hill 138 KV	Rebuild line with higher-capacity conductor	\$654,000	86% (\$561,000)
Bison Substation to Florida Avenue 138 KV	Reconductor line	\$909,000	73% (\$667,000)
	Total allocated to project		\$8,289,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.